Healthcare Utilization & Social Determinants:

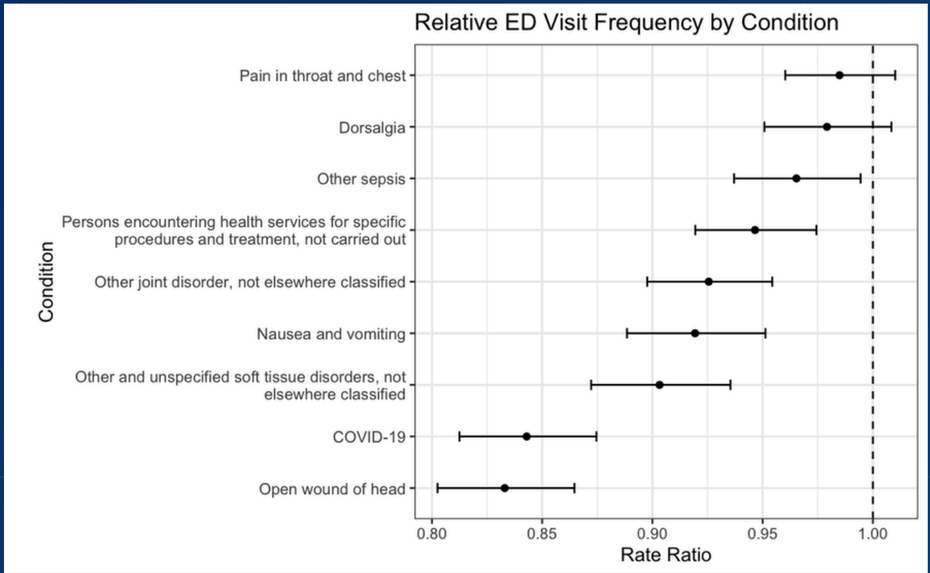
Trends in ED Use, Admissions, & Length of Stay

Apple seeRs

Hyunjin Lee, Liane Ma, Cooper Ruffing, Amy Xu



patient journeys



Frequent ED visitors (≥ 4) receive less consistent care

- See ~4× more providers
- Fewer repeat visits with same provider → episodic care

A small group drives disproportionate usage

- ~1.6% of patients are "frequent flyers"
- These patients account for much higher visit volume (~24.8%)

Category	Z-score	Adj p (FDR)
Alcohol Use	**8.490**	**0.000**
Physical Activity	**20.193**	**0.000**
Social Connections	**20.752**	**0.000**
Utilities	**9.947**	**0.000**
Financial Resource Strain	**5.534**	**0.000**
Transportation Needs	**5.213**	**0.000**
Food Insecurity	**3.963**	**0.000**
Housing Stability	1.345	0.112
Depression	-0.273	0.675
Intimate Partner Violence	-3.923	1.000

Higher ED Use with SDOH Risk

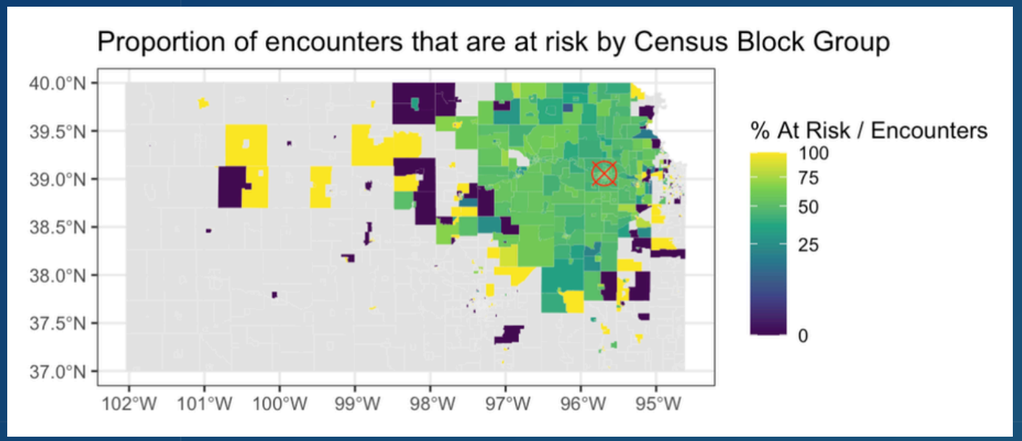
Patients w/ SDOH risk indicators were significantly different in **7/10 evaluated categories**

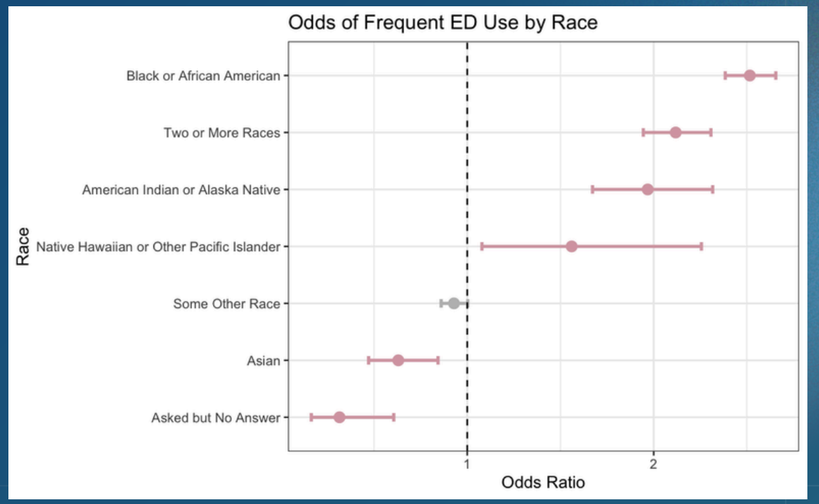
ED Visits are Episodic

ED visits are largely one-time and show little variation across conditions

At Risk ED encounters are focused in specific counties

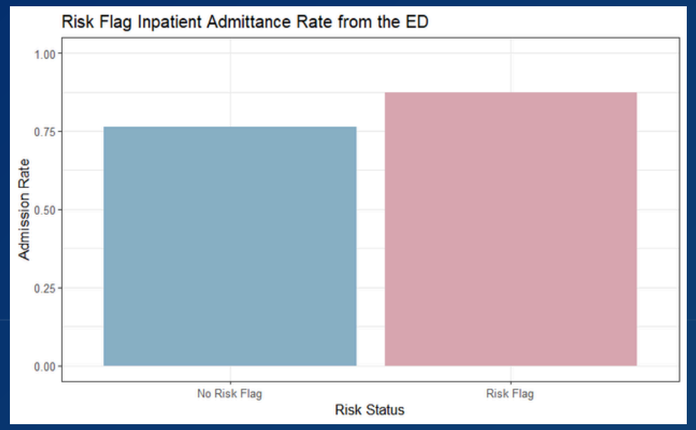
Focusing efforts on certain areas could help maximize the reach of healthcare





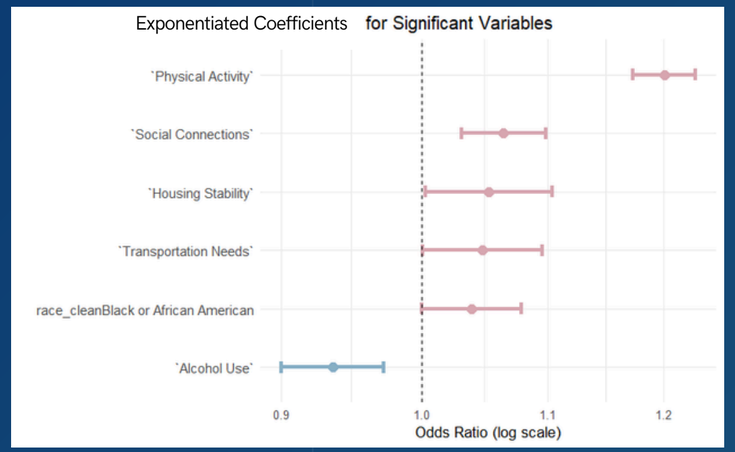
admission pathways

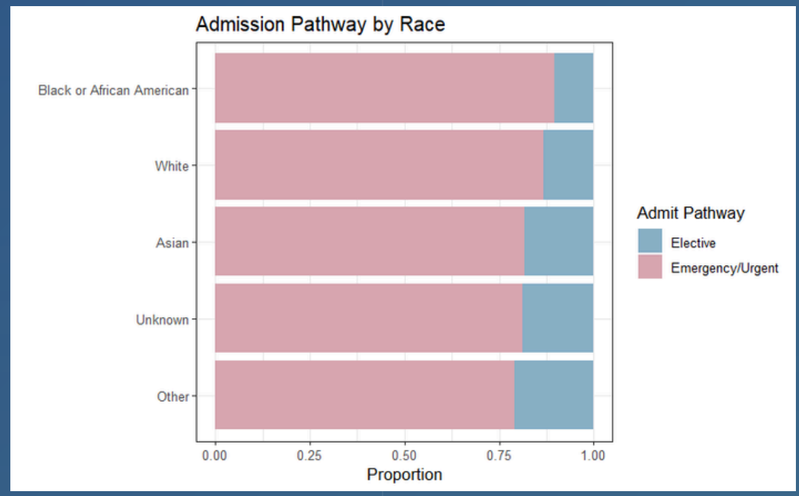
→ Are certain demographic groups more or less likely to enter via emergency/urgent/trauma pathways?



- SDOH Risk Flag significantly elevates inpatient admission odds from the ED

- SDOH Risk Indicators: lower physical activity, weaker social connections, housing instability, transportation needs predict higher LOS





Chi-Squared Test determined that admission pathway differs for both race and sex; different groups are admitted for different reasons

provider continuity

1. **Admission pathway is the main driver**
2. **Demographic factors**
3. **Behavioral factors: Alcohol Use**

Variable	Odds Ratio	p-value
`Alcohol Use`	1.17	0.037
raceBlack or African American	1.29	<0.001
sexOther/Unknown	0.62	<0.001
AdmissionTypeEmergency	3.01	<0.001
AdmissionTypeUrgent	2.09	<0.001

- Emergency Admissions - ~3x likely to switch providers (OR = 3.01)
- Urgent Admissions ~2x likely (OR = 2.09)
- Black patients ~29% higher odds
- Alcohol Use ~17% higher odds

recommendations

term <chr>	estimate <dbl>	p.value <dbl>
`Physical Activity`	0.18251992	1.446167e-50
`Social Connections`	0.06174158	1.460283e-04
`Alcohol Use`	-0.06662063	7.036805e-04
`Housing Stability`	0.05057192	3.804986e-02
`Transportation Needs`	0.04585221	4.519991e-02
race_cleanBlack or African American	0.03769090	4.892389e-02

term <chr>	coefficient <dbl>	odds_ratio <dbl>
`Transportation Needs`	0.57215549	1.77208265
`Physical Activity`	0.35288542	1.42316807
age_bin1970	0.26509766	1.30355828
race_cleanOther	0.22681461	1.25459726
age_bin1955	0.18761559	1.20636968
age_bin1945	0.13781421	1.14776228
`Social Connections`	0.10170048	1.10705183
race_cleanBlack or African American	0.07997036	1.08325496
`Housing Stability`	0.07502503	1.07791113
`Food Insecurity`	0.02682223	1.02718518

Linear Regression Model: Predicting Length of Stay

log(LOS) ~ emergency_flag + SDOH risk indicators + sex + age groups

- Lower physical activity, weaker social connections, housing instability, and transportation barriers are significantly associated with longer hospital stays. Alcohol use shows a small negative association with length of stay.

**emergency flag: "Emergency", "Urgent", "Trauma Center"*
No evidence of multicollinearity, no influential points, observed linear relationship

Lasso Regression: Predicting Frequent ED Use

frequent_flyer ~ SDOH risk indicators + sex + age groups + race

- Higher odds of frequent ED use: transportation bariers, lower levels of physical activity, weaker social connections, unstable housing situations, African-American/Black, food insecurity, middle age groups

Low multicollinearity, no problematic high leverage points, AUC = 0.67, MSE = .07

Future Implementations and Directions



1. **Expansion of Community Paramedicine/Mobile Integrated Healthcare**
    a. Currently - screenings, primary care services
    a. Future - assist with insurance, create plans to take meds/deliver prescriptions, patient transport
2. **Improve Social Connections**
    a. Connect with communities and groups, offer simple, free, low friction methods of entry
    a. Work with schools to ensure students are involved in their communities

Limitations: observational & survey data; results may not imply causation and may reflect differences in access and data completeness.